Malhar Srivatsav

Computer Science Undergraduate | Software Engineering, Backend & Application Security
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Education

Manipal Institute of Technology, Bangalore **2024 - Expected May 2028**
Bachelor of Technology in Computer Science *CGPA: 8.64/10*

Experience

Clazuno **June 2026 – Present**
Software Engineering Intern *Remote*

- Built a multi-step class creation wizard in React Native (Expo) with a schedule editor, time slot picker, and fee setup, integrated with existing REST APIs so trainers can create and publish classes end to end.
- Developed the home screen, day/week/month calendar views, and platform-adaptive business profile screens (native and web views, dynamic routing, refund policy modal).
- Set up the EAS Build pipeline for Android APK distribution and removed a deprecated Google Auth dependency, enabling staging builds for the team.
- Tested the web and mobile apps extensively and participated in code reviews and team meetings.

Rift Labs **2025 – Present**
Founding Engineer *Bangalore, India*

- Shipped **Singularity Vault**, a zero-knowledge offline Android password manager on Google Play, using AES-256-GCM with sub-2ms credential encrypt/decrypt latency.
- Achieved brute-force resistance at 1.15s key derivation per unlock using PBKDF2WithHmacSHA256 (100,000 iterations) without hurting everyday UX.
- Eliminated credential exposure with a 30-second idle timeout, 20-second clipboard auto-clear, and memory key wiping on backgrounding; zero cloud dependency.
- Layered Android BiometricPrompt over a master-password root of trust across 5 cryptographic operations.

Projects

Sentinel | *Go, Chi, PostgreSQL, Redis, Docker, GitHub Actions* | GitHub **2026**

- Built a deny-by-default RBAC authorization microservice in Go with 19 authenticated JSON endpoints, HS256 JWT auth, per-route scopes, strict input validation, and parameterized SQL across all 20 queries.
- Designed a Redis cache-aside layer with generation-based invalidation that fails safe to PostgreSQL on outages, an append-only audit log enforced by DB triggers, and per-IP and per-caller rate limiting.
- Wrote 570+ tests (89.9% internal coverage) and a GitHub Actions pipeline running the race detector, integration tests, govulncheck and gosec; it caught a reachable x/text vulnerability and a log-injection finding, both patched.

PhishGuard | *TypeScript, Node.js, Express, PostgreSQL* | GitHub **2026**

- Built a phishing detection REST API with 15+ URL risk signals (Shannon entropy, Punycode homographs, Damerau-Levenshtein typosquatting, double-encoding evasion) plus Google Safe Browsing and URLhaus feeds.
- Designed a normalized risk scoring engine with per-signal explainability in JSON, backed by PostgreSQL scan history, API key authentication, and rate limiting; deployed on Render.

Technical Skills

Languages: Python, Java, C, SQL, Go, Kotlin, JavaScript, TypeScript
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Testing
Frameworks & Mobile: React Native (Expo), React, Node.js, Express, Chi, Android (Jetpack Compose)
Backend, Data & Cloud: REST APIs, PostgreSQL, Redis, AWS (basic), Firebase (Auth, Firestore), Render
Engineering Practices: Git, GitHub, GitHub Actions (CI), Docker, Linux, Code Reviews, EAS Build
Security: JWT, RBAC, AES-256-GCM, PBKDF2, Android Keystore, BiometricPrompt, Rate Limiting, Web Application Security

Achievements & Certifications

- Solved 300+ data structures and algorithms problems on LeetCode.
- Finalist, Hack IITK 2026 – national cybersecurity hackathon hosted by IIT Kanpur's C3i Hub.
- JLPT N2 Certified (Japanese Language Proficiency), scored 156/180.